

Untitled

February 1, 2026

```
[1]: import pandas as pd
import numpy as np
import ast
import matplotlib.pyplot as plt

PATH = "aggregated_results.csv"
df = pd.read_csv(PATH)

# Keep only rows that are real amplicon features (Classification present)
df_feat = df[df["Classification"].notna()].copy()

# ---- helpers to parse list-like columns safely ----
def _parse_list_cell(x):
    if pd.isna(x):
        return []
    if isinstance(x, list):
        raw = x
    else:
        try:
            raw = ast.literal_eval(str(x))
        except Exception:
            return []
    if not isinstance(raw, list):
        return []
    out = []
    for item in raw:
        s = str(item).strip()
        s = s.strip("'").strip('"').strip()
        if s and s.lower() != "nan":
            out.append(s)
    return out

df_feat["oncogenes_list"] = df_feat["Oncogenes"].apply(_parse_list_cell)
df_feat["all_genes_list"] = df_feat["All genes"].apply(_parse_list_cell)

# numeric columns (coerce)
num_cols = [
```

```

    "Captured interval length",
    "Feature median copy number",
    "Feature maximum copy number",
    "Complexity score",
]
for c in num_cols:
    df_feat[c] = pd.to_numeric(df_feat[c], errors="coerce")

print("Rows:", df.shape, "Feature-rows:", df_feat.shape)
print("Classifications:", df_feat["Classification"].value_counts().to_dict())

```

Rows: (1234, 30) Feature-rows: (1157, 32)
Classifications: {'Linear': 550, 'ecDNA': 297, 'Complex-non-cyclic': 196, 'BFB': 114}

1 1)Is YES1 amplified as ecDNA or BFB in CCLE? In which samples?

```

[5]: gene = "YES1"

hits = df_feat[
    df_feat["oncogenes_list"].apply(lambda xs: gene in xs)
    | df_feat["all_genes_list"].apply(lambda xs: gene in xs)
].copy()

summary = (
    hits.groupby("Classification")["Sample name"]
    .nunique()
    .sort_values(ascending=False)
)

print("Unique samples with YES1 amplification by class:")
print(summary)

print("\nExample rows (sample, class, feature, locus):")
cols = ["Sample name", "Classification", "Feature ID", "Location",
        "Captured interval length", "Feature maximum copy number"]
print(hits[cols].sort_values(["Classification", "Sample name"]).head(30).
      to_string(index=False))

```

Unique samples with YES1 amplification by class:
Classification
BFB 1
Name: Sample name, dtype: int64

Example rows (sample, class, feature, locus):

Sample name	Classification	Feature ID
Location	Captured interval length	Feature maximum copy number
KYSE70_OESOPHAGUS	BFB KYSE70_OESOPHAGUS_amplicon7_BFB_1	
['chr18:634973-4361110']		3726137.0
21.688132		

2 2) Summarize the size distribution of all BFB amplifications (Captured interval length)

```
[9]: bfb = df_feat[df_feat["Classification"].eq("BFB")].copy()
      sizes = bfb["Captured interval length"].dropna()

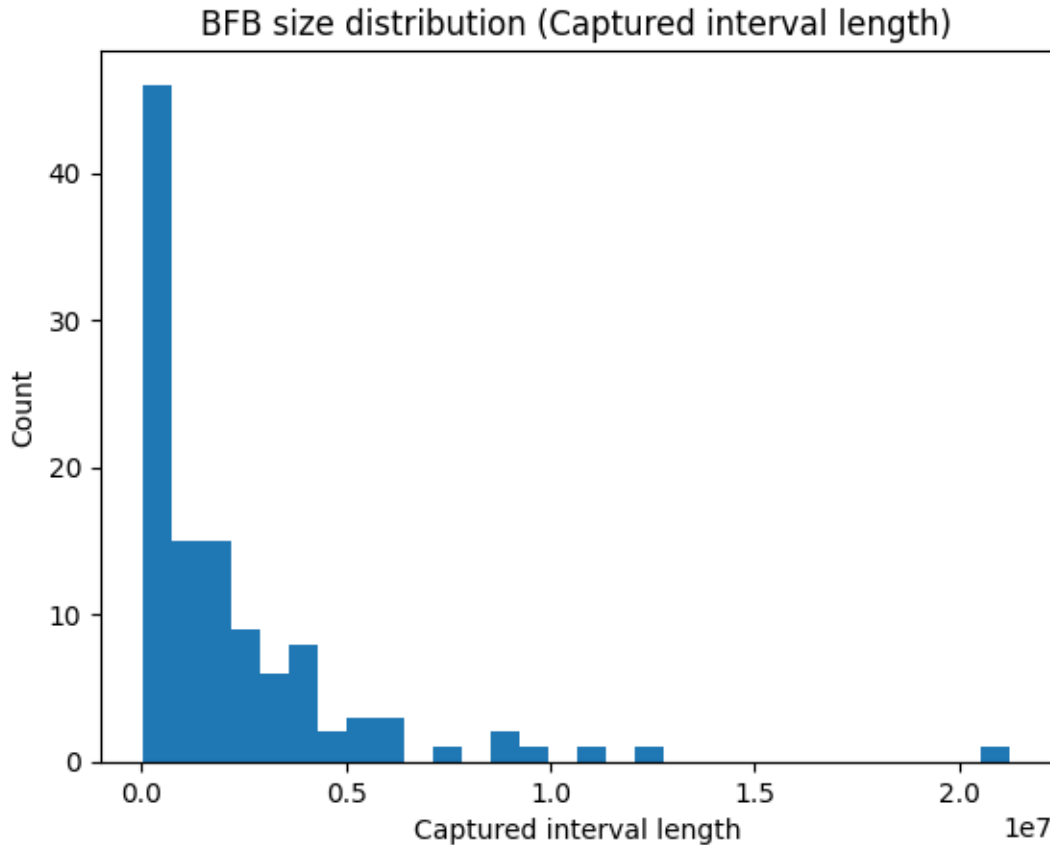
      print("BFB count:", len(bfb), "with size:", sizes.notna().sum())
      print(sizes.describe(percentiles=[0.1,0.25,0.5,0.75,0.9]))

      plt.figure()
      plt.hist(sizes, bins=30)
      plt.xlabel("Captured interval length")
      plt.ylabel("Count")
      plt.title("BFB size distribution (Captured interval length)")
      plt.show()
```

BFB count: 114 with size: 114

count	1.140000e+02
mean	2.271737e+06
std	2.991727e+06
min	5.946300e+04
10%	2.033848e+05
25%	5.037368e+05
50%	1.276367e+06
75%	2.929109e+06
90%	5.312285e+06
max	2.124683e+07

Name: Captured interval length, dtype: float64



3 3)List all BFB amplifications in HARA (feature IDs, loci, genes, copy number)

```
[11]: sample = "HARA_LUNG"
bt = df_feat[(df_feat["Sample name"] == sample) & (df_feat["Classification"] == "BFB")].copy()

cols = [
    "Sample name", "Classification", "AA amplicon number", "Feature ID", "Location",
    "Captured interval length", "Feature median copy number", "Feature maximum copy number",
    "Oncogenes", "All genes"
]
print("BFB features in", sample, ":", len(bt))
print(bt[cols].sort_values(["AA amplicon number", "Feature ID"]).to_string(index=False))
```

BFB features in HARA_LUNG : 1

Sample name	Classification	AA amplicon number	Feature ID
Location	Captured interval	length	Feature median copy number
copy number	Oncogenes		Feature maximum

All genes

HARA_LUNG	BFB	6.0 HARA_LUNG_amplicon6_BFB_1
["chr11:34893970-37730986", "chr11:37761005-41072281"]		
6148292.0	15.180671	56.560276
["APIP", "C11orf74", "CD44", "COMMD9", "FJX1", "LDLRAD3", "LRRC4C", "PAMR1", "PDHX", "PRR5L", "RAG1", "RAG2", "SLC1A2", "SNORD164", "TRAF6", "TRIM44"]		

4 4) Which genes are most frequently amplified as BFB? (top 25)

```
[12]: from collections import Counter

bfb = df_feat[df_feat["Classification"].eq("BFB")].copy()

ctr = Counter()
for genes in bfb["oncogenes_list"]:
    ctr.update(set(genes)) # set() to avoid double-counting within one feature

top = pd.DataFrame(ctr.most_common(25), columns=["Gene", "BFB_feature_count"])
print(top.to_string(index=False))
```

Gene	BFB_feature_count
CTTN	5
MYCL	4
FGF4	4
CCND1	4
FGF3	4
TRAF6	3
YAP1	3
MMP12	3
BIRC2	3
BIRC3	3
MIEN1	3
CDK12	3
ERBB2	3
LDHB	2
CCNE1	2
URI1	2
ACTN4	2
AKT2	2
CRKL	2
MYC	2
PVT1	2

KDR	2
KIT	2
MLLT6	2
LASP1	2

5 5) Show the distribution of amplification classes across cancer types (Tissue of origin)

```
[13]: tab = pd.crosstab(df_feat["Tissue of origin"], df_feat["Classification"]).
      ↪sort_index()
print(tab.head(20))

# Also show proportions per tissue
prop = tab.div(tab.sum(axis=1), axis=0).fillna(0)
print("\nProportions (first 20 tissues):")
print(prop.head(20))

# Plot top tissues by total feature count
top_tissues = tab.sum(axis=1).sort_values(ascending=False).head(15).index
plt.figure()
prop.loc[top_tissues].plot(kind="bar", stacked=True)
plt.ylabel("Proportion of features")
plt.title("Amplification class proportions by tissue (top 15 tissues)")
plt.tight_layout()
plt.show()
```

Classification	BFB	Complex-non-cyclic	Linear	ecDNA
Tissue of origin				
autonomic ganglia	0	0	0	1
bone	1	0	1	0
breast	13	32	101	62
central nervous system	0	5	7	6
endometrium	2	5	14	12
haematopoietic and lymphoid tissue	0	5	18	5
kidney	2	0	3	2
large intestine	3	7	12	4
liver	4	8	8	5
lung	43	67	193	104
oesophagus	11	12	40	21
ovary	7	13	59	28
pancreas	8	2	12	4
pleura	0	3	0	1
prostate	0	2	5	4
skin	2	4	13	2
soft tissue	0	1	1	0

stomach	5	14	35	27
thyroid	0	1	2	3
upper aerodigestive tract	8	9	19	0

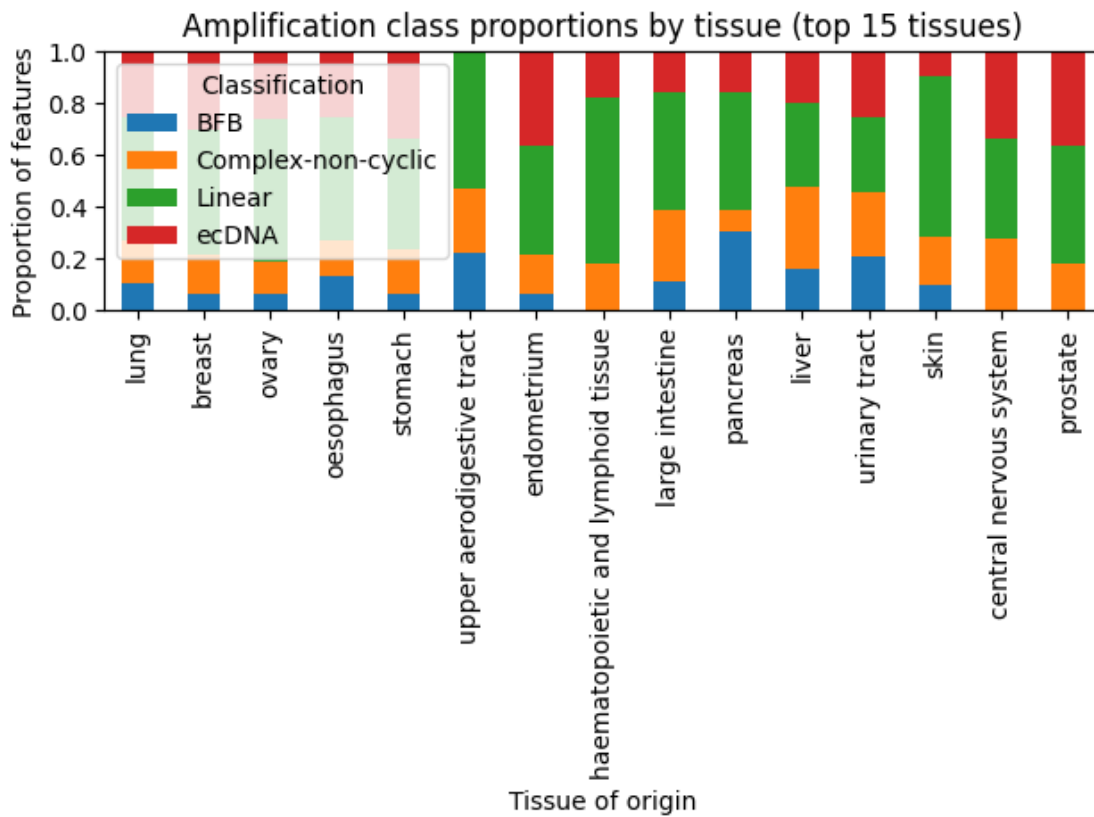
Proportions (first 20 tissues):

Classification	BFB	Complex-non-cyclic	Linear	\
Tissue of origin				
autonomic ganglia	0.000000	0.000000	0.000000	
bone	0.500000	0.000000	0.500000	
breast	0.062500	0.153846	0.485577	
central nervous system	0.000000	0.277778	0.388889	
endometrium	0.060606	0.151515	0.424242	
haematopoietic and lymphoid tissue	0.000000	0.178571	0.642857	
kidney	0.285714	0.000000	0.428571	
large intestine	0.115385	0.269231	0.461538	
liver	0.160000	0.320000	0.320000	
lung	0.105651	0.164619	0.474201	
oesophagus	0.130952	0.142857	0.476190	
ovary	0.065421	0.121495	0.551402	
pancreas	0.307692	0.076923	0.461538	
pleura	0.000000	0.750000	0.000000	
prostate	0.000000	0.181818	0.454545	
skin	0.095238	0.190476	0.619048	
soft tissue	0.000000	0.500000	0.500000	
stomach	0.061728	0.172840	0.432099	
thyroid	0.000000	0.166667	0.333333	
upper aerodigestive tract	0.222222	0.250000	0.527778	

Classification	ecDNA
Tissue of origin	
autonomic ganglia	1.000000
bone	0.000000
breast	0.298077
central nervous system	0.333333
endometrium	0.363636
haematopoietic and lymphoid tissue	0.178571
kidney	0.285714
large intestine	0.153846
liver	0.200000
lung	0.255528
oesophagus	0.250000
ovary	0.261682
pancreas	0.153846
pleura	0.250000
prostate	0.363636
skin	0.095238
soft tissue	0.000000
stomach	0.333333

```
thyroid                                0.500000
upper aerodigestive tract              0.000000
```

<Figure size 640x480 with 0 Axes>



6 6) For each tissue, what fraction of samples have any ecDNA?

```
[14]: # sample-level: does a sample have >=1 ecDNA feature?
sample_has_ec = (
    df_feat.assign(has_ecDNA=df_feat["Classification"].eq("ecDNA"))
    .groupby(["Tissue of origin", "Sample name"])["has_ecDNA"]
    .any()
    .reset_index()
)

tissue_frac = (
    sample_has_ec.groupby("Tissue of origin")["has_ecDNA"]
    .mean()
    .sort_values(ascending=False)
```



```

)

print("Top 25 tissues by ecDNA-sample fraction:")
print(tissue_frac.head(25))

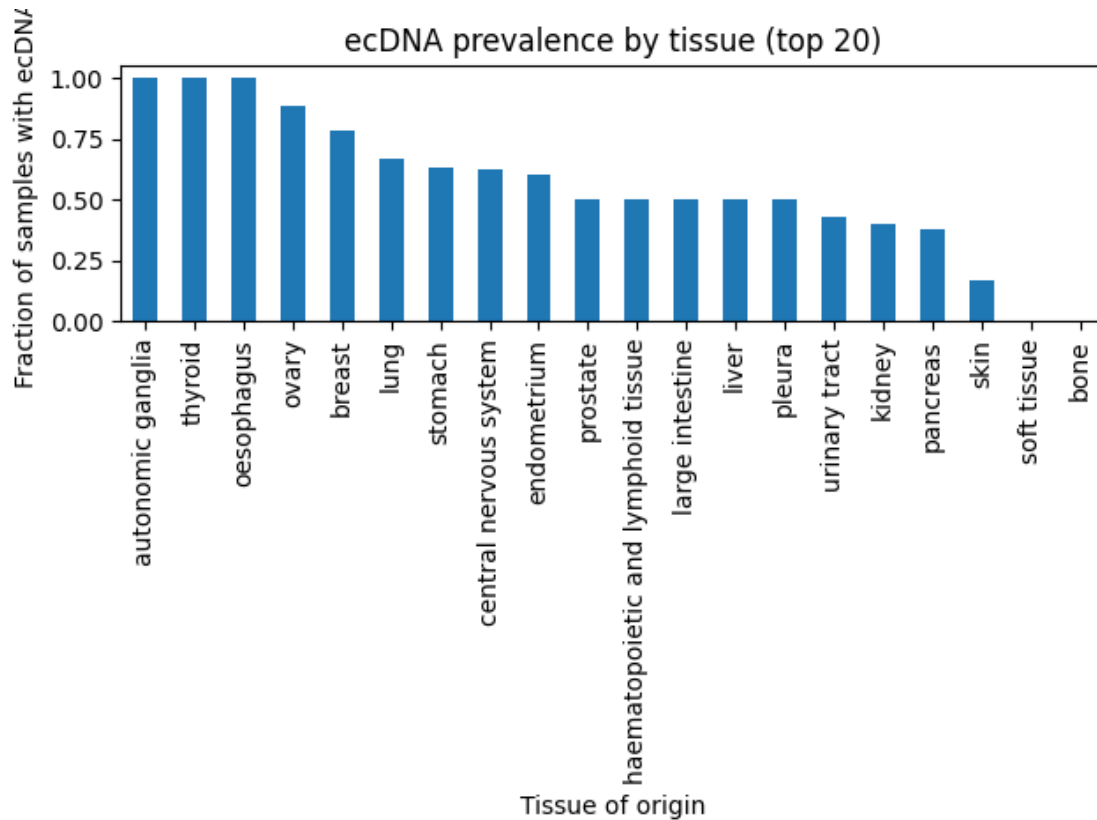
plt.figure()
tissue_frac.head(20).plot(kind="bar")
plt.ylabel("Fraction of samples with ecDNA")
plt.title("ecDNA prevalence by tissue (top 20)")
plt.tight_layout()
plt.show()

```

Top 25 tissues by ecDNA-sample fraction:

Tissue of origin	
autonomic ganglia	1.000000
thyroid	1.000000
oesophagus	1.000000
ovary	0.882353
breast	0.787879
lung	0.670455
stomach	0.631579
central nervous system	0.625000
endometrium	0.600000
prostate	0.500000
haematopoietic and lymphoid tissue	0.500000
large intestine	0.500000
liver	0.500000
pleura	0.500000
urinary tract	0.428571
kidney	0.400000
pancreas	0.375000
skin	0.166667
soft tissue	0.000000
bone	0.000000
upper aerodigestive tract	0.000000

Name: has_ecDNA, dtype: float64



7 7) Do Complexity scores differ between ecDNA vs BFB vs Linear vs Complex-non-cyclic?

```
[15]: x = df_feat[["Classification", "Complexity score"]].dropna().copy()

stats = x.groupby("Classification")["Complexity score"].describe(percentiles=[0.
    ↳ 25, 0.5, 0.75])
print(stats)

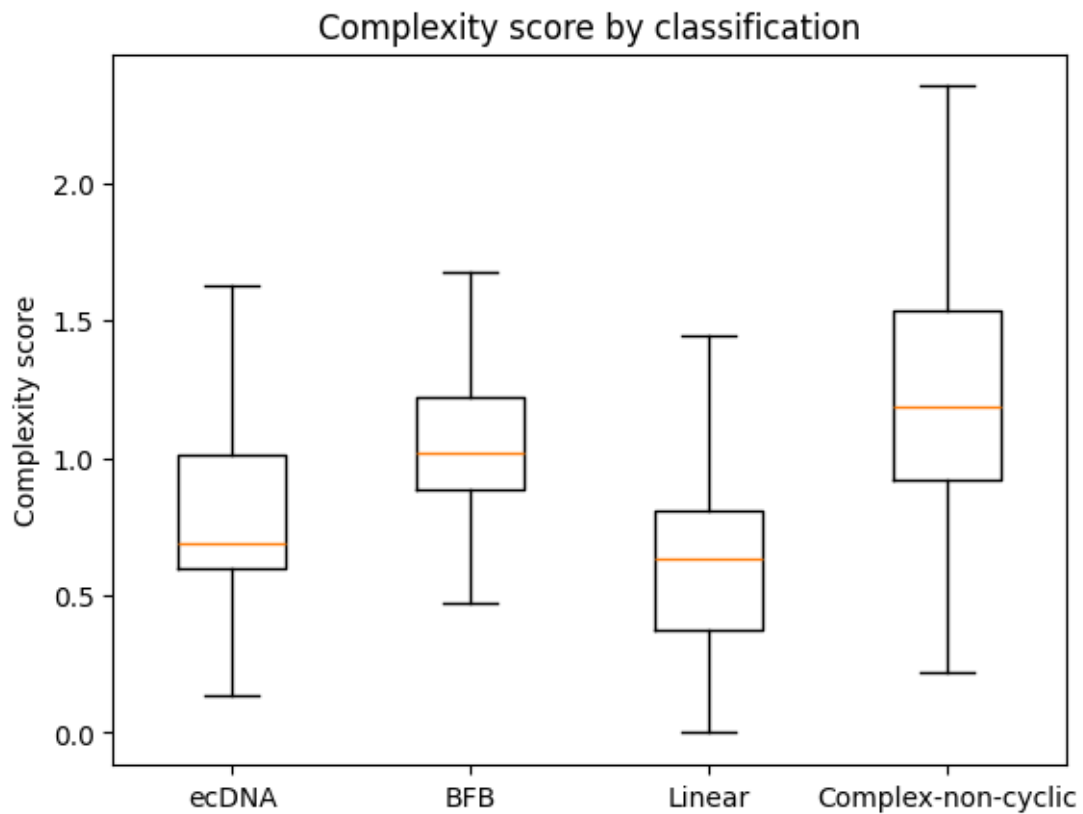
# simple boxplot (matplotlib)
classes = ["ecDNA", "BFB", "Linear", "Complex-non-cyclic"]
data = [x.loc[x["Classification"].eq(c), "Complexity score"].values for c in
    ↳ classes if c in x["Classification"].unique()]

plt.figure()
plt.boxplot(data, tick_labels=[c for c in classes if c in x["Classification"].
    ↳ unique()], showfliers=False)
plt.ylabel("Complexity score")
```

```
plt.title("Complexity score by classification")
plt.show()
```

	count	mean	std	min	25%	50%	\
Classification							
BFB	114.0	1.063654	0.256257	0.474421	0.887858	1.021014	
Complex-non-cyclic	196.0	1.261700	0.448434	0.222816	0.921391	1.189101	
Linear	550.0	0.580827	0.366532	0.000000	0.376822	0.636480	
ecDNA	297.0	0.840714	0.394129	0.133345	0.601398	0.686838	

	75%	max
Classification		
BFB	1.218679	1.758271
Complex-non-cyclic	1.533952	2.660781
Linear	0.810626	1.845296
ecDNA	1.012470	2.357014



8 8) For ecDNA vs BFB, how does size relate to max copy number? (scatter + correlation)

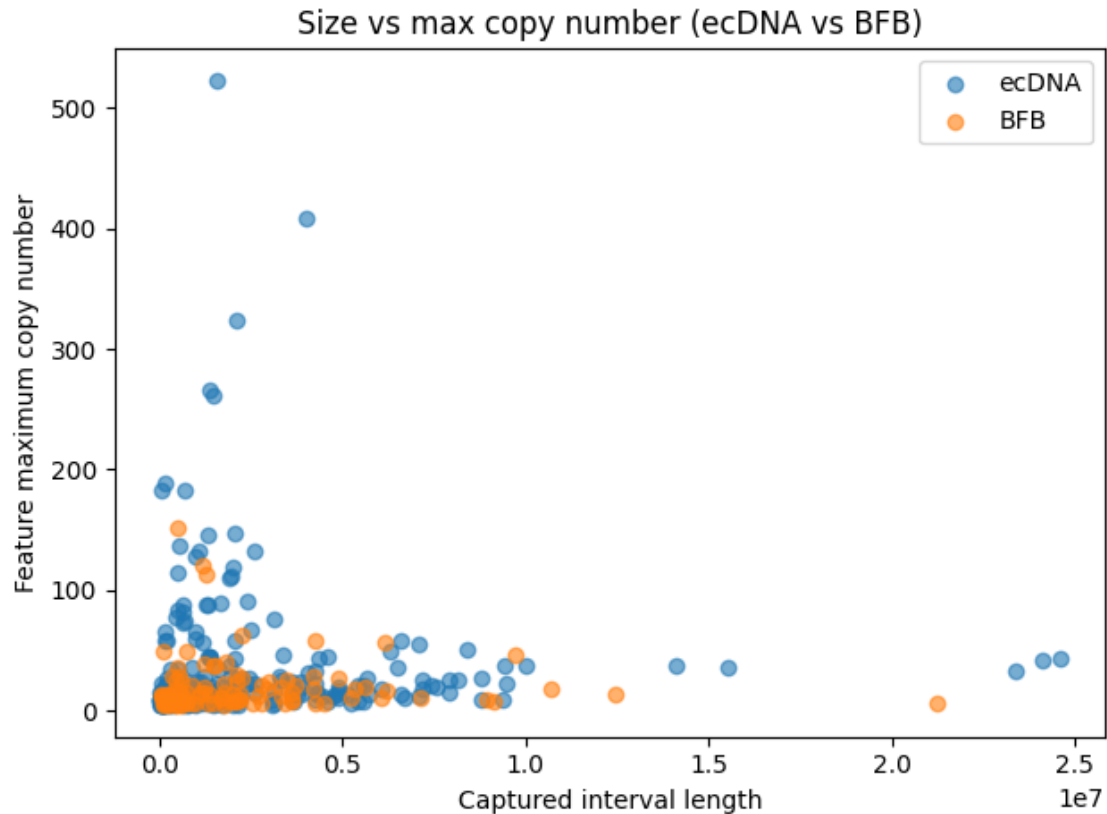
```
[16]: sub = df_feat[df_feat["Classification"].isin(["ecDNA","BFB"])] .copy()
sub = sub.dropna(subset=["Captured interval length","Feature maximum copy_
↪number"])

# correlations per class
for cls in ["ecDNA","BFB"]:
    s = sub[sub["Classification"] == cls]
    corr = s["Captured interval length"].corr(s["Feature maximum copy number"])
    print(f"{cls}: n={len(s)} corr(size, maxCN)={corr:.4f}")

# scatter
plt.figure()
for cls in ["ecDNA","BFB"]:
    s = sub[sub["Classification"] == cls]
    plt.scatter(s["Captured interval length"], s["Feature maximum copy_
↪number"], label=cls, alpha=0.6)
plt.xlabel("Captured interval length")
plt.ylabel("Feature maximum copy number")
plt.title("Size vs max copy number (ecDNA vs BFB)")
plt.legend()
plt.tight_layout()
plt.show()
```

ecDNA: n=297 corr(size, maxCN)=0.0498

BFB: n=114 corr(size, maxCN)=-0.0029



9 9) What are the highest copy-number amplifications in CCLE? (top 5)

```
[18]: top = (
    df_feat.dropna(subset=["Feature maximum copy number"])
            .sort_values("Feature maximum copy number", ascending=False)
            .head(5)
)

cols = ["Sample name", "Tissue of origin", "Classification", "Feature_
↳ ID", "Location",
        "Feature maximum copy number", "Feature median copy number", "Captured_
↳ interval length", "Oncogenes"]
print(top[cols].to_string(index=False))
```

Sample name	Tissue of origin	Classification	Feature ID	Location	Feature maximum copy number	Feature median copy number	Captured interval length	Oncogenes

interval	length			Oncogenes
NCIH524_LUNG	lung	ecDNA		
NCIH524_LUNG_amplicon2_ecDNA_1				
["'chr8:120528251-120544068'", "'chr8:125645887-126514397'",				
"'chr8:127229750-127914137'"]		522.977215		
220.331345	1568714.0			
["'MYC'", "'PVT1'"]				
SNU16_STOMACH	stomach	ecDNA		
SNU16_STOMACH_amplicon1_ecDNA_1				
["'chr10:120319822-120638334'",				
"'chr10:120819706-121603030'", "'chr10:121603879-121900616'",				
"'chr11:34748870-36277002'", "'chr11:37963818-38181183'",				
"'chr11:41437269-41437834'", "'chr18:51191728-51191838'",				
"'chr8:98244213-98244504'", "'chr8:126509826-126580205'",				
"'chr8:126600763-126600906'", "'chr8:127142839-127143050'",				
"'chr8:127222146-128043192'"]		408.881192		
90.103116	4036815.0			["'FGFR2'",
"'MYC'", "'PVT1'"]				
NCIH2170_LUNG	lung	ecDNA		
NCIH2170_LUNG_amplicon2_ecDNA_1				
["'chr8:101178106-101178974'", "'chr8:126935950-127072035'",				
"'chr8:127338657-128452923'", "'chr8:128459019-128947060'",				
"'chr17:39661313-39808423'", "'chr17:39809908-39883401'",				
"'chr17:40151306-40259508'", "'chr17:40259581-40311678'"]				
323.685194	233.393630	2120162.0		["'CDC6'",
"'ERBB2'", "'MIEN1'", "'MYC'", "'PVT1'", "'RARA'"]				
KATOIII_STOMACH	stomach	ecDNA		
KATOIII_STOMACH_amplicon1_ecDNA_2				
["'chr10:120316892-120467177'", "'chr10:120469265-121715975'"]				
265.719353	11.087396	1396995.0		
["'FGFR2'"]				
NCIH69_LUNG	lung	ecDNA		
NCIH69_LUNG_amplicon1_ecDNA_1				
["'chr2:15911407-16027706'",				
"'chr2:16032497-16048750'", "'chr2:52932410-52964250'",				
"'chr2:52977759-53034483'", "'chr2:53052137-53306581'",				
"'chr2:53320409-53689668'", "'chr2:53703524-53732658'",				
"'chr2:76985056-76993446'", "'chr2:86504308-86507094'",				
"'chr2:134519878-134657863'", "'chr2:136821917-136844590'",				
"'chr2:139750547-140138583'", "'chr2:145130284-145181197'"]				
262.219802	150.432204	1484736.0		
["'MYCN'"]				

10 10) Which oncogenes appear in both ecDNA and BFB across CCLE?

```
[20]: from collections import defaultdict

gene_classes = defaultdict(set)

for _, r in df_feat.iterrows():
    for g in r["oncogenes_list"]:
        gene_classes[g].add(r["Classification"])

both = sorted([g for g, cls in gene_classes.items() if {"ecDNA", "BFB"} <= cls])

print("Genes appearing in both ecDNA and BFB:")
print(both)
print("Count:", len(both))
```

Genes appearing in both ecDNA and BFB:

```
['ARID2', 'ARNT', 'BIRC2', 'BIRC3', 'BRF2', 'CAV1', 'CCND1', 'CCND3', 'CCNE1',
'CDK12', 'CHIC2', 'COPS3', 'CSF3', 'CSF3R', 'CTTN', 'EGFR', 'EMSY', 'ERBB2',
'FGF3', 'FGF4', 'FIP1L1', 'FOXA1', 'FOXG1', 'GAB2', 'INTS3', 'INTS4', 'KDR',
'KIT', 'KRAS', 'LASP1', 'LDHB', 'MCL1', 'MDM2', 'MECOM', 'MET', 'MIEN1',
'MLANA', 'MLLT11', 'MLLT6', 'MMP12', 'MPL', 'MYC', 'MYCL', 'NKX2-1', 'PAK1',
'PDGFRA', 'PRKCI', 'PRMT5', 'PVT1', 'RPL23', 'RSF1', 'SEC62', 'SETDB1', 'SFPQ',
'SKIL', 'SMURF1', 'SREBF1', 'TFEB', 'THRA', 'THRAP3', 'TRAF6', 'TRIO', 'URI1',
'WNT2', 'YAP1', 'YEATS4', 'ZNF703']
```

Count: 67

11 12) Which samples contain both ecDNA and BFB amplifications?

```
[23]: sample_classes = (
    df_feat.groupby("Sample name")["Classification"]
    .apply(set)
)

both = sample_classes[sample_classes.apply(lambda x: {"ecDNA", "BFB"} <= x)]
print("Samples with both ecDNA and BFB:")
print(both.index.tolist())
print("Count:", len(both))
```

Samples with both ecDNA and BFB:

```
['ABC1_LUNG', 'AU565_BREAST', 'BT20_BREAST', 'CHAGOK1_LUNG', 'CORL88_LUNG',
'COV644_OVARY', 'DANG_PANCREAS', 'DMS114_LUNG', 'DMS273_LUNG', 'EBC1_LUNG',
'EFM19_BREAST', 'HARA_LUNG', 'HCC1500_BREAST', 'HCC1599_BREAST',
```

```
'HCC1954_BREAST', 'HCC38_BREAST', 'HCC827_LUNG', 'HCC95_LUNG', 'JHH7_LIVER',
'KLE_ENDOMETRIUM', 'KYSE180_OESOPHAGUS', 'KYSE30_OESOPHAGUS',
'KYSE410_OESOPHAGUS', 'KYSE70_OESOPHAGUS', 'MDAMB157_BREAST',
'MDAMB175VII_BREAST', 'MDAMB453_BREAST', 'MKN1_STOMACH', 'MKN74_STOMACH',
'NCIH1092_LUNG', 'NCIH1435_LUNG', 'NCIH1651_LUNG', 'NCIH1694_LUNG',
'NCIH1792_LUNG', 'NCIH1876_LUNG', 'NCIH2029_LUNG', 'NCIH2066_LUNG',
'NCIH2170_LUNG', 'NCIH2171_LUNG', 'NCIH3255_LUNG', 'NCIH520_LUNG',
'NCIH650_LUNG', 'NCIH661_LUNG', 'NCIH889_LUNG', 'NIHOVCAR3_OVARY',
'OE19_OESOPHAGUS', 'OV90_OVARY', 'RMGI_OVARY', 'SCABER_URINARY_TRACT',
'SNU182_LIVER', 'TE11_OESOPHAGUS']
```

Count: 51

12 13) Are ecDNA amplifications larger than BFB on average?

```
[25]: sizes = (
    df_feat[df_feat["Classification"].isin(["ecDNA","BFB"])]
    .groupby("Classification")["Captured interval length"]
    .describe()
)

print(sizes)
```

	count	mean	std	min	25% \
Classification					
BFB	114.0	2.271737e+06	2.991727e+06	59463.0	503736.75
ecDNA	297.0	2.096639e+06	3.299880e+06	10387.0	260748.00

	50%	75%	max
Classification			
BFB	1276367.0	2929108.75	21246834.0
ecDNA	985738.0	2233311.00	24593843.0

13 14) Which samples have high-copy (>20 CN) ecDNA?

```
[5]: hc = df_feat[
    (df_feat["Classification"] == "ecDNA") &
    (df_feat["Feature maximum copy number"] > 20)
]

print(hc[["Sample name","Location","Feature maximum copy number","Oncogenes"]])
```

	Sample name \
2	5637_URINARY_TRACT
5	59M_OVARY

12	8305C_THYROID
35	AU565_BREAST
55	BT474_BREAST
...	...
1148	SNU475_LIVER
1155	SW1783_CENTRAL_NERVOUS_SYSTEM
1199	THP1_HAEMATOPOIETIC_AND_LYMPHOID_TISSUE
1212	UACC812_BREAST
1219	UACC893_BREAST

	Location \
2	['chr6:20350615-22839372']
5	['chr8:52712046-56014576', 'chr8:65533018-6...']
12	['chr20:34283685-34285398', 'chr20:34299250...']
35	['chr8:71754758-71754831', 'chr8:71755523-7...']
55	['chr20:44404279-45225559', 'chr20:49762732...']
...	...
1148	['chr11:66696384-66698194', 'chr11:66702589...']
1155	['chr4:5688943-5689095', 'chr4:53913454-568...']
1199	['chr1:39491061-40894493', 'chr2:241579699-...']
1212	['chr13:48317568-48959208', 'chr13:48962485...']
1219	['chr17:39329205-40331868', 'chr17:66007324...']

	Feature maximum copy number \
2	67.419329
5	24.854069
12	44.561896
35	33.639403
55	36.933944
...	...
1148	20.047833
1155	24.354231
1199	42.827031
1212	35.315340
1219	56.898700

	Oncogenes
2	['E2F3', 'SOX4']
5	['TCEA1']
12	['DUSP26', 'PIGU']
35	['CCN3', 'HEY1', 'MYC', 'PVT1', 'TPD5...']
55	['AURKA', 'BMP7', 'CYP24A1', 'RAB22A', ...]
...	...
1148	['CCND1', 'CTTN', 'FGF3', 'FGF4', 'KD...']
1155	['CHIC2', 'KDR', 'KIT', 'PDGFRA']
1199	['MYCL']
1212	['BMP7', 'CDC6', 'CDK12', 'CSF3', 'ER...']
1219	['CDC6', 'CDK12', 'CSF3', 'ERBB2', 'M...']

[98 rows x 4 columns]

14 15) What is the largest amplicon (by size) per class?

```
[6]: largest = (
    df_feat.sort_values("Captured interval length", ascending=False)
    .groupby("Classification")
    .head(1)
)

cols = ["Classification", "Sample name", "Location", "Captured interval_
↪length", "Oncogenes"]
print(largest[cols].to_string(index=False))
```

Classification	Sample name
Location	Captured interval length
Oncogenes	
Complex-non-cyclic	SNU119_OVARY
["'chr8:115269237-125582885'", "'chr8:125588887-126179965'", 'chr8:126182327-126394148'", "'chr8:126397480-128452922'", 'chr8:128459020-131255992'", "'chr8:131268907-137730724'", 'chr8:137731618-139960606'", "'chr8:139962905-144831650'", 'chr8:144832447-145076005'"]	
	29772069.0
["'CCN3'", "'CCN4'", "'EEF1D'", "'EXT1'", "'MAFA'", "'MYC'", "'NDRG1'", 'PTP4A3'", "'PVT1'", "'RAD21'", "'RECQL4'", "'TRIB1'"]	
ecDNA	HCC1419_BREAST
["'chr8:81134388-83146029'", "'chr8:87281194-87776995'", 'chr8:89137435-89140200'", "'chr8:89140228-89141640'", 'chr8:89141881-89143076'", "'chr8:89489148-89489289'", 'chr8:89495519-89499492'", "'chr8:89684982-91939500'", 'chr8:92174442-93060099'", "'chr8:93064964-93645323'", 'chr8:117833541-119142305'", "'chr8:119149088-122101275'", 'chr8:122102832-122808352'", "'chr8:123010220-125582885'", 'chr8:125588890-126179961'", "'chr8:126182327-128452923'", 'chr8:128459019-130395930'", "'chr8:139003367-139370467'", 'chr8:139397262-139401535'", "'chr8:140319540-140452381'", 'chr8:143853259-143853365'", "'chr20:42069079-42072282'", 'chr20:44675621-44710394'", "'chr20:49448996-49802965'", 'chr20:50243820-51182240'", "'chr20:51544314-51572841'", 'chr20:53152709-53715421'", "'chr20:58629246-59464368'", 'chr20:62288512-62288514'", "'chr20:62662833-62671291'", 'chr20:62671407-62672846'", "'chr20:63631155-63636888'", 'chr20:63637081-63640776'", "'chr20:63641828-63643034'", 'chr20:63644875-63964865'", "'chr20:63966082-64045660'", 'chr17:39530968-39798137'", "'chr17:48664668-49274433'"]	

```

"chr17:49275687-50414463"]                24593843.0 ["CCN3", "CDK12",
"COL1A1", "CTSZ", "ERBB2", "EXT1", "GNAS", "IGF2BP1", "ITGA3",
"MIEN1", "MYC", "PVT1", "TRIB1", "ZNF217"]
                BFB                HEPG2_LIVER
["chr14:22006768-26121753", "chr14:26123186-29258843",
"chr14:29259596-32484096", "chr14:32485156-34959590",
"chr14:34963248-39629107", "chr14:39631009-39848113",
"chr14:39848192-40140611", "chr14:40148468-40907546",
"chr14:40907558-41139967", "chr14:41200503-42847941",
"chr14:42848945-43331896"]                21246834.0
["FOXA1", "FOXG1", "NKX2-1", "PRMT5"]
                Linear SCC25_UPPER_AERODIGESTIVE_TRACT
["chr1:87780712-89010224", "chr1:89012924-91668677",
"chr1:91669311-99225964"]                11441918.0
["EVI5", "GFI1", "RPL5"]

```

15 16) Which tissues show highest amplification complexity (median)?

```

[7]: tissue_complexity = (
    df_feat.groupby("Tissue of origin")["Complexity score"]
    .median()
    .sort_values(ascending=False)
)

print(tissue_complexity.head(20))

```

Tissue of origin	
pleura	1.248873
autonomic ganglia	1.176241
soft tissue	1.129679
liver	0.973848
large intestine	0.961082
kidney	0.872199
urinary tract	0.869917
pancreas	0.840527
skin	0.810184
upper aerodigestive tract	0.807988
stomach	0.788245
central nervous system	0.759224
lung	0.719535
ovary	0.691793
oesophagus	0.688151
haematopoietic and lymphoid tissue	0.681421
breast	0.681368
endometrium	0.676187

```
prostate          0.664281
thyroid           0.591204
Name: Complexity score, dtype: float64
```

16 17) For a given gene, list all co-amplified genes (same feature)

```
[8]: gene = "EGFR"

co = set()
for _, r in df_feat.iterrows():
    if gene in r["all_genes_list"]:
        co |= set(r["all_genes_list"])

co.discard(gene)
print("Co-amplified genes with", gene)
print(sorted(co))
```

Co-amplified genes with EGFR

```
['CALN1', 'CCT6A', 'COBL', 'DDC', 'DDC-AS1', 'EGFR-AS1', 'ELDR', 'FIGNL1',
'FKBP9P1', 'GRB10', 'IKZF1', 'LANCL2', 'MRPS17', 'NIPSNAP2', 'POM121L12',
'PSPH', 'SEC61G', 'SEPTIN14', 'SPATA48', 'UBE2J2', 'VOPP1', 'VSTM2A',
'VSTM2A-OT1', 'ZNF713', 'ZPBP']
```

17 18) Which genes are exclusive to ecDNA (never BFB or Linear)?

```
[9]: gene_cls = defaultdict(set)

for _, r in df_feat.iterrows():
    for g in r["oncogenes_list"]:
        gene_cls[g].add(r["Classification"])

exclusive = sorted([g for g,c in gene_cls.items() if c == {"ecDNA"}])
print("ecDNA-exclusive genes:", exclusive)
```

ecDNA-exclusive genes: ['AHI1', 'AKTIP', 'ARAF', 'ATM', 'BAALC', 'BCAR4',
'BCL7A', 'BMI1', 'CAD', 'CARD11', 'CBLL1', 'CD24', 'CIITA', 'COPS5', 'COX6C',
'CSNK2A1', 'CTNNB1', 'DDHD2', 'DDX10', 'DDX6', 'DUSP26', 'EIF3E', 'ELK1',
'FGFR2', 'FUS', 'FZD2', 'GNA12', 'GOPC', 'HEY1', 'HMGA2', 'IFNG', 'IGF2BP1',
'IKZF1', 'JUN', 'JUP', 'KMT2A', 'KSR2', 'LSM1', 'MALT1', 'MAML2', 'MAP2K4',
'MAP3K8', 'MNAT1', 'MSI2', 'MTDH', 'MYB', 'MYCN', 'NANOG', 'NCOA3', 'NME1',
'OTX2', 'PAX8', 'PBX1', 'PICALM', 'RAD51B', 'RARA', 'RB1', 'RMI2', 'RNF217-AS1',
'RNF43', 'ROS1', 'RUNX1T1', 'SALL4', 'SGK1', 'SIX1', 'SOCS1', 'SS18L1', 'SUFU',

```
'TACSTD2', 'TBC1D15', 'TCEA1', 'TPD52', 'TREH', 'TRIB2', 'TRIM8', 'UBR5',  
'VTI1A', 'WWTR1']
```

18 19) Which chromosomes are most frequently amplified?

```
[10]: chroms = (  
    df_feat["Location"]  
    .str.extract(r"(chr\d+|chrX|chrY)", expand=False)  
)  
  
chrom_counts = chroms.value_counts()  
print(chrom_counts)
```

```
Location  
chr1      122  
chr8      104  
chr11     101  
chr3       77  
chr12      74  
chr6       62  
chr17      62  
chr7       55  
chr2       54  
chr20      48  
chr5       44  
chr14      42  
chr19      38  
chr16      34  
chrX       34  
chr4       33  
chr9       31  
chr10      31  
chr22      29  
chr18      25  
chr13      24  
chr15      21  
chr21      11  
Name: count, dtype: int64
```

19 20) Which oncogenes are most recurrent across unique samples (any class)?

```
[11]: pairs = []
      for _, r in df_feat.iterrows():
          for g in set(r["oncogenes_list"]):
              pairs.append((g, r["Sample name"]))

      pairs = pd.DataFrame(pairs, columns=["Gene", "Sample"])

      top = (
          pairs.drop_duplicates()
              .groupby("Gene")["Sample"]
              .nunique()
              .sort_values(ascending=False)
      )

      print(top.head(30))
```

Gene	
PVT1	33
MYC	33
CTTN	23
FGF4	23
CCND1	23
FGF3	23
MIEN1	19
ERBB2	19
MYCL	17
CDK12	15
KDM2A	12
URI1	11
KRAS	11
MECOM	10
CRKL	10
MUC4	9
FGFR1	8
CCNE1	8
YAP1	8
BRF2	8
NSD3	8
SEC62	7
MMP12	7
BIRC3	7
BIRC2	7
NKX2-1	7
CHIC2	7

```

ZNF703      7
KAT6B       6
GAB2        6
Name: Sample, dtype: int64

```

20 21) For each tissue, what are the top 5 amplified oncogenes (any class, by #unique samples)?

```

[12]: rows = []
      for _, r in df_feat.iterrows():
          t = r["Tissue of origin"]
          s = r["Sample name"]
          for g in set(r["oncogenes_list"]):
              rows.append((t, g, s))

      x = pd.DataFrame(rows, columns=["Tissue", "Gene", "Sample"]).drop_duplicates()

      top5 = (
          x.groupby(["Tissue", "Gene"])["Sample"].nunique()
            .reset_index(name="n_samples")
            .sort_values(["Tissue", "n_samples"], ascending=[True, False])
            .groupby("Tissue")
            .head(5)
      )

      print(top5.to_string(index=False))

```

	Tissue	Gene	n_samples
	autonomic ganglia	MYCN	1
	autonomic ganglia	TRIB2	1
	bone	RECQL4	1
	breast	ERBB2	10
	breast	MIEN1	10
	breast	CDK12	8
	breast	PVT1	7
	breast	BRF2	6
	central nervous system	CHIC2	2
	central nervous system	KDR	2
	central nervous system	KIT	2
	central nervous system	PDGFRA	2
	central nervous system	CCND1	1
	endometrium	ACTN4	1
	endometrium	AKT2	1
	endometrium	CCNE1	1
	endometrium	CDK12	1
	endometrium	CEBPA	1

haematopoietic and lymphoid tissue	ETV3	2
haematopoietic and lymphoid tissue	NTRK1	2
haematopoietic and lymphoid tissue	PRCC	2
haematopoietic and lymphoid tissue	AGAP2	1
haematopoietic and lymphoid tissue	ARNT	1
kidney	CCND3	1
kidney	CD274	1
kidney	MLANA	1
kidney	TFEB	1
large intestine	AHI1	1
large intestine	ALDH1A1	1
large intestine	CCND3	1
large intestine	COL1A1	1
large intestine	CYGB	1
liver	CCND1	3
liver	CTTN	3
liver	FGF3	3
liver	FGF4	3
liver	COPS3	2
lung	MYC	18
lung	MYCL	16
lung	PVT1	15
lung	BCL9	4
lung	CHD1L	4
oesophagus	CTTN	4
oesophagus	CCND1	3
oesophagus	FGF3	3
oesophagus	FGF4	3
oesophagus	BIRC2	2
ovary	MECOM	4
ovary	FGFR3	3
ovary	FGFR4	3
ovary	MUC4	3
ovary	MYC	3
pancreas	ACTN4	1
pancreas	AGAP2	1
pancreas	AKT2	1
pancreas	ALDH1A1	1
pancreas	BRF2	1
pleura	MYC	1
pleura	PVT1	1
prostate	CDK1	1
prostate	KAT6B	1
prostate	NKX2-1	1
prostate	RAD51B	1
skin	CLTCL1	2
skin	CRKL	2
skin	INPPL1	2

skin	NUMA1	2
skin	BIRC2	1
soft tissue	KRAS	1
stomach	MYC	4
stomach	PVT1	4
stomach	KRAS	3
stomach	CAV1	2
stomach	CDK12	2
thyroid	AIMP2	1
thyroid	CHIC2	1
thyroid	DUSP26	1
thyroid	FIP1L1	1
thyroid	KDR	1
upper aerodigestive tract	CTTN	5
upper aerodigestive tract	CCND1	4
upper aerodigestive tract	FGF3	4
upper aerodigestive tract	FGF4	4
upper aerodigestive tract	INPPL1	2
urinary tract	CCND1	2
urinary tract	CTTN	2
urinary tract	E2F3	2
urinary tract	FGF3	2
urinary tract	FGF4	2

21 22) Which tissues have the highest fraction of BFB features?

```
[13]: tab = pd.crosstab(df_feat["Tissue of origin"], df_feat["Classification"])
frac_bfb = (tab["BFB"] / tab.sum(axis=1)).sort_values(ascending=False)
print(frac_bfb.head(25))
```

Tissue of origin	
bone	0.500000
pancreas	0.307692
kidney	0.285714
upper aerodigestive tract	0.222222
urinary tract	0.208333
liver	0.160000
oesophagus	0.130952
large intestine	0.115385
lung	0.105651
skin	0.095238
ovary	0.065421
breast	0.062500
stomach	0.061728
endometrium	0.060606
haematopoietic and lymphoid tissue	0.000000

central nervous system	0.000000
pleura	0.000000
prostate	0.000000
soft tissue	0.000000
thyroid	0.000000
autonomic ganglia	0.000000
dtype: float64	

22 23) Among samples with ecDNA, what are the most common co-amplified oncogene pairs?

```
[14]: from itertools import combinations
      from collections import Counter

      ec = df_feat[df_feat["Classification"]=="ecDNA"]

      pair_ctr = Counter()
      for xs in ec["oncogenes_list"]:
          genes = sorted(set(xs))
          for a,b in combinations(genes, 2):
              pair_ctr[(a,b)] += 1

      top_pairs = pd.DataFrame(
          [(*k, v) for k,v in pair_ctr.most_common(30)],
          columns=["Gene1", "Gene2", "ecDNA_feature_count"]
      )
      print(top_pairs.to_string(index=False))
```

Gene1	Gene2	ecDNA_feature_count
MYC	PVT1	23
ERBB2	MIEN1	13
CDK12	ERBB2	10
CDK12	MIEN1	10
CCND1	FGF3	10
CCND1	FGF4	10
FGF3	FGF4	10
CCND1	CTTN	9
CTTN	FGF3	9
CTTN	FGF4	9
BRF2	ZNF703	6
BRF2	DDHD2	6
BRF2	FGFR1	6
BRF2	LSM1	6
BRF2	NSD3	6
DDHD2	FGFR1	6
DDHD2	LSM1	6

DDHD2	NSD3	6
FGFR1	LSM1	6
FGFR1	NSD3	6
LSM1	NSD3	6
DDHD2	ZNF703	5
FGFR1	ZNF703	5
LSM1	ZNF703	5
NSD3	ZNF703	5
CDK12	CSF3	4
CDK12	THRA	4
CSF3	ERBB2	4
CSF3	MIEN1	4
CSF3	THRA	4

23 24) Which samples have amplifications on multiple chromosomes? (by feature loci)

```
[15]: chrom = df_feat["Location"].str.extract(r"(chr\d+|chrX|chrY)", expand=False)
tmp = df_feat.copy()
tmp["chrom"] = chrom

per_sample_chr = (
    tmp.dropna(subset=["chrom"])
    .groupby("Sample name")["chrom"]
    .nunique()
    .sort_values(ascending=False)
)

print("Top 25 samples by #distinct amplified chromosomes:")
print(per_sample_chr.head(25))
```

Top 25 samples by #distinct amplified chromosomes:

Sample name	
NCIH1651_LUNG	13
KYSE30_OESOPHAGUS	11
OVSARO_OVARY	11
MFE280_ENDOMETRIUM	10
MKN7_STOMACH	10
NCIH2066_LUNG	10
NCIH1792_LUNG	9
EBC1_LUNG	9
HCC1569_BREAST	8
NCIH23_LUNG	8
FUOV1_OVARY	8
FADU_UPPER_AERODIGESTIVE_TRACT	8
HARA_LUNG	8

```

BT20_BREAST      8
BT474_BREAST     7
HCC38_BREAST     7
EFM192A_BREAST   7
KYSE410_OESOPHAGUS 7
HCC95_LUNG       7
SKCO1_LARGE_INTESTINE 7
NIHOVCAR3_OVARY  7
SHP77_LUNG       6
OVCAR8_OVARY     6
RMGI_OVARY       6
HDQP1_BREAST     6
Name: chrom, dtype: int64

```

24 25) For each class, what is the distribution of number of oncogenes per feature?

```

[16]: df_feat["oncogene_count"] = df_feat["oncogenes_list"].apply(lambda xs: len(set(xs)))

stats = (
    df_feat.groupby("Classification")["oncogene_count"]
    .describe(percentiles=[0.25,0.5,0.75,0.9])
)
print(stats)

# optional plot
classes = sorted(df_feat["Classification"].dropna().unique())
data = [df_feat.loc[df_feat["Classification"].eq(c), "oncogene_count"].values
        for c in classes]

plt.figure()
plt.boxplot(data, tick_labels=classes, showfliers=False)
plt.ylabel("# oncogenes per feature")
plt.title("Oncogene count per feature by class")
plt.xticks(rotation=30, ha="right")
plt.tight_layout()
plt.show()

```

	count	mean	std	min	25%	50%	75%	90%	max
Classification									
BFB	114.0	1.412281	1.871119	0.0	0.0	1.0	2.0	4.0	10.0
Complex-non-cyclic	196.0	1.897959	2.784535	0.0	0.0	1.0	3.0	6.0	14.0
Linear	550.0	0.532727	1.320230	0.0	0.0	0.0	1.0	2.0	17.0
ecDNA	297.0	1.727273	2.499263	0.0	0.0	1.0	2.0	5.0	17.0

